

## LAREDO, TX, USA

WMO: 722520

Lat: 27.533N

Lon: 99.467W

Elev: 151

StdP: 99.53

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12907

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 2.2        | 3.9 | -10.0                         | 1.6 | 11.8 | -7.5 | 2.0 | 12.3 | 11.9                  | 18.7 | 10.9 | 18.0 | 3.8                      | 0    | 0.530 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.1                   | 40.6            | 23.7 | 39.4 | 23.7 | 38.4 | 23.7 | 26.4                | 33.3 | 26.0 | 32.8 | 25.6 | 32.5 | 5.6                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.0                            | 20.5 | 28.2 | 24.4 | 19.7 | 28.1 | 24.0 | 19.2 | 28.1 | 83.0          | 33.2 | 81.2 | 32.8 | 79.7 | 32.8 | 30.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 11.0              | 9.7  | 8.9 | DB                         | -0.9 | 42.5               | 1.8 | 1.2                                                | -2.3 | 43.4       | -3.3 | 44.1       | -4.3 | 44.8       | -5.7 | 45.7 |  |
|                   |      |     | WB                         | -2.8 | 27.7               | 1.5 | 1.2                                                | -3.8 | 28.5       | -4.7 | 29.2       | -5.5 | 29.9       | -6.6 | 30.7 |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 24.1   | 14.8  | 17.4  | 20.9  | 25.0  | 28.5  | 31.0  | 31.5  | 31.6  | 28.7  | 24.7  | 19.4  | 15.1  |
|                                                                   | DBStd       | 7.29   | 5.10  | 5.47  | 4.98  | 3.93  | 3.10  | 2.08  | 1.73  | 1.87  | 2.75  | 4.14  | 4.94  | 5.06  |
|                                                                   | HDD10.0     | 42     | 17    | 8     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 12    |
|                                                                   | HDD18.3     | 427    | 131   | 77    | 33    | 4     | 0     | 0     | 0     | 0     | 0     | 7     | 47    | 128   |
|                                                                   | CDD10.0     | 5178   | 166   | 214   | 339   | 450   | 573   | 631   | 665   | 671   | 560   | 457   | 283   | 169   |
|                                                                   | CDD18.3     | 2523   | 22    | 49    | 112   | 204   | 315   | 381   | 407   | 413   | 310   | 205   | 78    | 27    |
|                                                                   | CDH23.3     | 30839  | 170   | 401   | 982   | 2087  | 3799  | 5261  | 5752  | 5874  | 3694  | 2054  | 585   | 178   |
|                                                                   | CDH26.7     | 16795  | 40    | 132   | 399   | 1013  | 2053  | 3117  | 3453  | 3563  | 1914  | 891   | 179   | 39    |
| Wind                                                              | WSAvg       | 4.9    | 4.1   | 4.4   | 5.1   | 5.5   | 5.8   | 5.8   | 5.7   | 5.2   | 4.2   | 4.4   | 4.1   | 3.9   |
| Precipitation                                                     | PrecAvg     | 498    | 25    | 20    | 30    | 36    | 62    | 44    | 62    | 40    | 84    | 43    | 28    | 23    |
|                                                                   | PrecMax     | 736    | 101   | 64    | 109   | 106   | 130   | 158   | 218   | 212   | 177   | 147   | 78    | 107   |
|                                                                   | PrecMin     | 293    | 1     | 0     | 0     | 2     | 2     | 0     | 0     | 0     | 14    | 0     | 0     | 2     |
|                                                                   | PrecStd     | 114    | 27    | 19    | 29    | 29    | 36    | 41    | 55    | 54    | 44    | 41    | 27    | 30    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 30.3  | 33.4  | 35.8  | 39.0  | 41.1  | 42.0  | 41.4  | 41.4  | 39.9  | 36.5  | 32.6  | 30.4  |
|                                                                   |             | MCWB   | 18.4  | 18.6  | 19.4  | 21.7  | 24.1  | 24.0  | 23.8  | 23.7  | 23.0  | 24.0  | 22.0  | 19.5  |
|                                                                   | 2%          | DB     | 27.5  | 30.3  | 32.9  | 36.3  | 38.6  | 40.1  | 40.0  | 40.2  | 37.7  | 34.4  | 31.0  | 27.6  |
|                                                                   |             | MCWB   | 17.4  | 18.8  | 19.8  | 21.3  | 23.5  | 24.1  | 23.6  | 23.8  | 23.7  | 23.3  | 21.4  | 18.4  |
|                                                                   | 5%          | DB     | 25.1  | 27.9  | 31.1  | 34.1  | 37.1  | 38.7  | 38.9  | 39.0  | 36.2  | 32.9  | 28.7  | 25.3  |
|                                                                   |             | MCWB   | 16.4  | 18.2  | 19.8  | 21.1  | 23.5  | 24.2  | 23.5  | 23.9  | 23.6  | 22.9  | 20.2  | 17.8  |
|                                                                   | 10%         | DB     | 22.7  | 25.9  | 28.8  | 32.4  | 35.1  | 37.4  | 37.7  | 37.8  | 34.6  | 31.5  | 27.0  | 22.9  |
|                                                                   |             | MCWB   | 15.8  | 17.4  | 19.4  | 20.8  | 23.4  | 23.9  | 23.4  | 23.8  | 23.4  | 22.4  | 19.5  | 17.4  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 21.1  | 21.9  | 23.3  | 25.0  | 26.6  | 26.9  | 26.9  | 26.7  | 26.7  | 26.2  | 24.0  | 21.9  |
|                                                                   |             | MCDB   | 26.2  | 27.9  | 29.8  | 31.7  | 35.4  | 35.0  | 32.4  | 32.7  | 32.3  | 31.7  | 28.9  | 26.6  |
|                                                                   | 2%          | WB     | 19.7  | 20.9  | 22.3  | 24.1  | 25.8  | 26.2  | 26.2  | 26.0  | 25.8  | 25.4  | 23.0  | 20.9  |
|                                                                   |             | MCDB   | 23.8  | 26.3  | 28.4  | 30.9  | 33.7  | 34.2  | 32.5  | 32.9  | 31.3  | 30.7  | 27.6  | 24.4  |
|                                                                   | 5%          | WB     | 18.6  | 20.1  | 21.6  | 23.5  | 25.2  | 25.7  | 25.7  | 25.7  | 25.5  | 24.7  | 22.0  | 19.9  |
|                                                                   |             | MCDB   | 22.5  | 24.9  | 27.4  | 30.1  | 32.6  | 33.4  | 32.3  | 32.7  | 31.0  | 29.8  | 26.5  | 23.2  |
|                                                                   | 10%         | WB     | 17.5  | 19.2  | 20.8  | 22.8  | 24.7  | 25.3  | 25.3  | 25.3  | 25.1  | 24.0  | 21.1  | 18.6  |
|                                                                   |             | MCDB   | 21.2  | 23.5  | 26.0  | 29.1  | 31.6  | 32.7  | 32.1  | 32.4  | 30.5  | 28.9  | 25.3  | 22.3  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.4  | 11.7  | 11.7  | 12.4  | 12.0  | 12.3  | 12.2  | 12.1  | 10.8  | 11.0  | 10.8  | 11.0  |
|                                                                   |             | MCDBR  | 14.3  | 14.3  | 14.3  | 14.5  | 14.4  | 14.0  | 14.0  | 13.6  | 12.6  | 11.8  | 12.4  | 13.3  |
|                                                                   |             | MCWBR  | 6.9   | 5.9   | 5.3   | 5.0   | 3.8   | 2.8   | 2.7   | 2.7   | 3.1   | 3.7   | 5.0   | 6.5   |
|                                                                   | 5% WB       | MCDBR  | 11.0  | 11.4  | 11.6  | 12.2  | 12.2  | 12.4  | 11.9  | 12.1  | 10.5  | 10.1  | 10.2  | 10.4  |
|                                                                   |             | MCWBR  | 6.2   | 5.8   | 4.9   | 4.9   | 3.7   | 3.1   | 3.0   | 3.0   | 3.1   | 3.9   | 5.0   | 6.3   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.325 | 0.341 | 0.364 | 0.390 | 0.412 | 0.404 | 0.398 | 0.397 | 0.396 | 0.360 | 0.348 | 0.325 |
|                                                                   | taud        |        | 2.517 | 2.473 | 2.418 | 2.365 | 2.348 | 2.426 | 2.445 | 2.465 | 2.424 | 2.517 | 2.484 | 2.526 |
|                                                                   | Ebn at Noon |        | 924   | 934   | 927   | 908   | 882   | 883   | 888   | 889   | 883   | 902   | 889   | 908   |
|                                                                   | Edh at Noon |        | 91    | 103   | 115   | 125   | 127   | 117   | 114   | 111   | 113   | 97    | 93    | 87    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 3.20  | 3.92  | 4.89  | 5.71  | 6.08  | 6.91  | 6.83  | 6.63  | 5.37  | 4.56  | 3.51  | 2.90  |
|                                                                   | RadStd      |        | 0.40  | 0.66  | 0.61  | 0.55  | 0.56  | 0.33  | 0.51  | 0.32  | 0.47  | 0.53  | 0.42  | 0.47  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | +0.51 | N/A     | N/A    | +0.86   | N/A   | +0.35 | N/A         | N/A     | +184    | +161    |
| (47) | Regional (15 neighbors) | +0.35 | N/A     | N/A    | +0.50   | N/A   | N/A   | N/A         | N/A     | +134    | +117    |

Nomenclature: See separate page